



DAY 26: JAVASCRIPT PROTOTYPES & PROTOTYPE CHAIN



Ever wondered how JavaScript objects inherit properties?
 The answer lies in Prototypes!

◆ What is a Prototype in JavaScript?

Every JavaScript object has an internal link to another object called its prototype. This prototype object can have its own prototype, forming a prototype chain.

✓ Think of it like inheritance – if an object doesn't have a property, JavaScript looks up the prototype chain to find it.

◆ Creating an Object with a Prototype

```
const person = {  
  greet: function () {  
    console.log("Hello!");  
  },  
};  
const user = Object.create(person); //  
'user' inherits from 'person'  
user.greet(); // Output: "Hello!"
```

- ◆ The user object does not have a greet method, but JavaScript finds it in its prototype (person).

◆ Understanding the Prototype Chain

```
const animal = {  
  eat: function () {  
    console.log("Eating...");  
  },  
};  
  
const dog = Object.create(animal);  
dog.bark = function () {  
  console.log("Woof! Woof!");  
};  
  
console.log(dog.eat()); // Inherited from  
                          'animal'  
console.log(dog.bark()); // Defined in  
                          'dog'
```

🔧 How JavaScript finds a property?

- 1 It looks in the dog object.
- 2 If not found, it looks in dog's prototype (animal).
- 3 If still not found, it moves further up the prototype chain (until reaching Object.prototype).

◆ Prototype vs. `__proto__` vs. `prototype` 🤯

🚀 `__proto__` (Deprecated, but still in use)

Every JavaScript object has a `__proto__` property that points to its prototype.

```
console.log(dog.__proto__ === animal);  
           // true
```

prototype (Used in Constructor Functions & Classes)

The prototype property exists on functions and is used when creating objects with new.

Example:

```
function Car(name) {  
  this.name = name;  
}
```

```
Car.prototype.drive = function () {  
  console.log("Driving...");  
};
```

```
const myCar = new Car("Tesla");  
myCar.drive(); // "Driving..."
```

- ◆ Objects created with new inherit from Car.prototype.

- ◆ Checking Object's Prototype

- ✓ Using Object.getPrototypeOf():

```
console.log(Object.getPrototypeOf  
  (dog) === animal); // true
```

- ✓ Using instanceof:

```
console.log(myCar instanceof Car); // true
```


◆ Overriding Prototype Properties

If an object has its own property, it overrides the prototype's property.

```
const person = {  
  greet: function () {  
    console.log("Hello!");  
  },  
};
```

```
const user = Object.create(person);  
user.greet = function () {  
  console.log("Hi there!");  
};
```

```
user.greet(); // "Hi there!" (Overrides  
prototype method)
```


◆ Prototype Chain in Built-in Objects

JavaScript's built-in objects (like Array, Object, Function) also have prototypes.

```
const arr = [1, 2, 3];  
console.log(arr.__proto__ ===  
Array.prototype); // true  
console.log(arr.__proto__.__proto__ ===  
Object.prototype); // true
```

- ◆ Arrays inherit from Array.prototype, which inherits from Object.prototype.